



FUEL HAND.- OPN Department
SIMHADRI

Note No :1



Ref:SMTTP/FUEL HAND.-/2023-24/LMI-CHP/446073

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Subject:Handling and Storage of Coal LMI/OGN/OPS/SYST/013

Sub: Updation of LMI-Handling and Storage of Coal "LMI/OGN/OPS/SYST/013"

'Appendix – VI: Belt Mapping Format' in LMI-Handling and Storage of Coal "LMI/OGN/OPS/SYST/013" has been updated in accordance with OGN "COS-ISO-00-OGN/OPS/SYST/013", Appendix – VI.

Put-up for approval of competent authority for implementation.

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Revised Appendix VI is in line with OGN/COS. May be approved.

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Location Management Instructions

LMI/OGN/OPS/SYST/013

Issue No.1, Revision No.4, Sept'2023

LMI

Handling & Storage of Coal

Approved for
Implementation by

.....
Sanjay Kumar Sinha

HOP - Simhadri

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

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I. INTRODUCTION:

Simhadri STPP sources 8 to 10 million metric tons of Indian Coal, mainly from Mahanadi Coal Fields (MCL) through FSA, and from other sources like ECL, SCCL and CCL as per requirement. Operational efficiency of plant is profoundly influenced by quality of coal.

Coal from mines is dispatched through railway rakes either through BOBR wagons or BOXN/S wagons. The coal supplied through BOBR rakes will be unloaded at Track Hoppers (TH) i.e TH-1 or TH-2 and the Coal supplied through BOXN/S wagons will be unloaded at Wagon tippers (WT) i.e WT-1 or WT-2. After unloading the coal is either stacked in stock piles or directed to bunkers as per requirement after crushing in respective Crusher houses.

This LMI Handling & Storage of Coal is updated and deals with guidelines for operation & maintenance of coal handling plant, coal stock yard management, proper blending of coal and monsoon preparation action plan for Coal Handling Plants. Various guidelines/circulars issued from time to time and recommended practices are compiled in the form of comprehensive manual.

II. SUPERCEDED DOCUMENTS:

- a. SUPPLY, SAMPLING, ANALYSIS, HANDLING & STORAGE OF COAL OF CHP
LMI/OGN/OPS/SYST/026
- b. HANDLING & STORAGE OF COAL, LMI/OGN/OPS/SYST/013, REVISION:2

III. RECORDS OF REVISION:

This LMI has been revised totally in accordance with the new OGN issued by CC-OS, "COS-ISO-00-OGN/OPS/SYST/013" Rev. No.: 2, June 2022.

IV. SECTION-IA : OPERATION & MAINTENANCE OF COAL HANDLING PLANT

1.0 PREAMBLE

This LMI specifies the requirements for Operating and Maintaining the Coal Handling Plant Effectively, Economically and Safely.

Simhadri Coal Handling plant comprises of Unloading facilities and associated equipment (i.e THs, WTs and PFs), Conveyor network supported with trestles and galleries, crushing equipment (Crushers & VGFs), various transfer points, Mobile trippers, Stacker cum reclaimers, cooling water systems etc. The coal is unloaded at Wagon Tipplers or Track Hoppers and is conveyed to bunkers or stock-yard. With so much of structural, moving and rotating equipment it is necessary to carefully consider the operation & maintenance of the plant.

OPERATION

General Instructions

Before starting the plant, check which of the equipment is available and accordingly plan an appropriate path for the movement of coal

Also, before starting any plant, check that the equipment is not under P.T.W.

Ensure that the path so planned, is clear for coal movement i.e. Flap Gate position, Interlocking etc.

Check from each location that the equipment is ready for start and no persons or material are hindering the equipment.

Line-up the system by starting each equipment from the last conveyor in the up-stream path, i.e. from bunkering path or stacking path as applicable.

Good communication between the control room and different equipment locations is essential. The Shift-charge-engineer shall check the communication system and ensure that communication between Control Room and different locations is available.

For better equipment performance and safe operation, it is essential to ensure that the equipment is operating satisfactorily while it is in operation. This can be achieved by checking each part of the system as per the illustrative points mentioned at [Appendix-I](#).

All sites like Wagon tipplers, Stacker Reclaimers, Bunkers, Track hoppers, Crushers wherever Operators are posted for monitoring in shift must have LOGSHEETS/LOGBOOKS monitoring the critical parameters, equipment status and activities as per station configuration. All Operators in the locations shall fill-in these Log sheets as per format of stations. (Soft Copy may be used).

Based on the operation checklist and walk down check list (As per *illustrative points mentioned in Appendix-I*), the Operation Engineer will issue work orders/notification to maintenance section identifying defects which require rectification.

CHP operation person looking after housekeeping must take PTW (likewise maintenance) from shift operation for works like cleaning of VGF/Crusher/Stacker reclaimers or clearing of any overloading/choking which requires working near movable equipment.

Executives shall carryout walk down inspection of the plant and maintain a record of the same in a register/online. Station should make a schedule based on station configuration and exceptions/observations of walk down checks as per records shall be reviewed.

Station must comply OIN-OPS-SYST-075, Rev-0, Reliability Improvement Matrix of Coal Handling Plant by adopting as LMI.

CHP executives may suggest plant improvement as per format decided by station.

A change-over schedule is to be prepared and the equipment change-over is to be followed as per the schedule.

Shift charge engineer shall log the operation activities in a logbook or in E-Log covering following or more parameters as per station need:

- Bunker Levels. Rake receiving, unloaded, balance and in pipeline.
- Coal received, stacked, bunkered.
- Belt-Weigher, Running Hours and Energy Meter Readings (on day basis) to calculate Utilization factor and specific energy.
- Biomass and imported coal handled in shift (Fed in which mills with quantity).
- PTWs issued and returned.
- Equipment in distress or breakdown.
- Major activities carried out in shift, defects attended and important instruction to next shift.
- Any instructions of HOD to be followed in all groups.
- Any other parameter suitable for station.

The Shift-Charge-Engineer shall maintain the following documentation as per format enclosed at *Appendix-XI, XII & XIII*.

- a) PTW Register indicating the equipment for which the PTW is issued and cancelled in the shift.
- b) Equipment availability.
- c) Protection/Interlock registers.
- d) Communications register.

All Metal Detectors and Magnetic Separators are to be kept in service. Detectors/Separators should be inspected daily and foreign material should be

removed, where necessary. MDs to be tested on daily basis by standard test piece (Test Piece shall be as per Clause 4.5.3 & 4.5.4 of OGN COS-IS0-00-0TN/PERF/009 Rev. No.: I) and record to be maintained. Sandbag marker, counter of each MD to be maintained and detail of collected materials to be recorded. As per format in [Appendix-IX](#).

CHP operator shall monitor that sump pumps provided at various locations are operated regularly during the shift. Excess water should be pumped away, where necessary.

All Electrical/C&I systems should be healthy. Availability of protection systems requires special attention by Operation Engineers, and these are to be regularly monitored.

- a. Hooters
- b. Pull Cord switches
- c. Zero-speed switches
- d. Belt sway switches
- e. Chute block switches
- f. Crusher bearing vibration monitoring equipment
- g. Interlocks/Protections
- h. PA System
- i. Magnetic Separators
- j. Metal Detectors

By-passing of these protections is to be avoided. In case of any problems in any of the protection systems, these shall immediately be brought to the notice of the Maintenance Department. For bypassing of protection the procedures laid down in "Procedure for Regulating By-passing of permissive and protection" OD/OPS/SYST/005 should be followed.

Electro Magnetic/Thrust Brakes installed in long conveyors should be kept in good working order. If these are not working it may cause non synchronous stoppage of different belts leading to blockage of the transfer point chute.

Adequate lighting should be ensured in all the areas.

Frequent starts and stop should be avoided as far as possible and no unwarranted operation should be carried-out.

The General Operation Instructions for the major equipment is given in the subsequent paragraphs. Before taking any equipment into service the operating instructions are to be followed.

All Operation & Maintenance Engineers should read the O&M manual for the Coal Handling Plant and implement the content of the document.

Stone - Picking

Stone pieces contained in the coal from the mine-end are to be removed manually from running conveyors before the coal reaches the crusher house/bunker.

Proper platforms should be made for stone-pickers on either side of conveyor and they should use full body safety harness while performing this job. Minimum three persons per location shall be deployed in a row for effective and safe operation of stone picking. Arrangement of suitable lighting and PA system shall be provided at stone picking locations. Also, one independent pull cord switch (other than PCS of conveyor)/EPB which cannot be bypassed must be provided in vicinity of each stone picker through which they can trip the conveyor. More protections like on belt safety switches (Spring tension based) or infra-red based systems must also be deployed to have zero probability of accident in stone picker area.

Stones should be disposed of through specially fabricated chute openings in the walking platform. The stone disposing area below walking platform should be properly fenced / barricaded.

Paddle Feeders

Paddle Feeders feeds coal on to the conveyor belts from the Track hopper at a controlled rate. The paddle blades projected in the hopper slot scoop the coal from hoppers and discharge it on the conveyor belt running below.

The following points need to be checked before taking any paddle feeder to service.

- a) The paddle blades are free from any obstruction.
- b) There is no leakage at gear boxes and the oil levels are correct.
- c) There is no leakage of hydraulic oil (hydraulically driven paddle feeders), and the oil levels are correct.
- d) Cable-reeling-drums are correctly winding and unwinding.
- e) The paddle feeder is operating only in interlock mode with conveyor.
- f) Speed control mechanism of paddle feeder to be kept healthy.
- g) Anti-collision device if provided to be kept healthy.
- h) Belt stop arrangement from Paddle Feeder is healthy.
- i) The flap gate is in correct position.
- j) End stop switches are functioning at both ends.
- k) PA system for communication with CHP control room is available.
- l) Adequate lights at operating panel of PF

Paddle Feeders have provision for operating in local or remote. Remote Operation of paddle feeders to be avoided as the coal containing boulders/stones and foreign materials may damage the paddle feeders. They should only be operated in local mode.

Conveyors

Conveyors are provided across various TPs for transporting the coal from unloading points i.e Wagon-tipler/Track hopper to either Unit Bunkers or Stock yards.

The following points need to be checked before taking any conveyor into service:

- a) There is no oil leakage at gear boxes/fluid couplings and the oil levels are correct.
- b) The Discharge and Receiving chutes are free from jamming and choking.
- c) The Flap Gates are in correct positions.
- d) All inspection doors are in the closed position.
- e) All pull-cord, Zero Speed (Tripping Time settings to be reviewed), Belt sway, Chute block switches are in service and pull cord ropes are intact.
- f) The Alarm Siren/hooter is in working order.
- g) Communication facilities like PA system /Mobiles are available.
- h) The lighting is adequate.
- i) No PTW is pending.
- j) Manpower designated to monitor these conveyors like stone picker, housekeeping staff are available on site.
- k) Ensure availability of Dust Control Systems.

Crushers

Ring Granulator type crushers are provided to crush the coal from (-) 250 mm size to (-) 20 mm size.

The following points need to be checked before returning any crushers to service:

- a) The crusher shall be clear from any jamming.
- b) All inspection doors are in the closed position and locked.
- c) There is no leakage of oil from Fluid couplings and the oil levels are correct.
- d) All Hammers & Cage bars are intact.
- e) ZSS switches are in service.
- f) Internals should be intact and there should not be any hanging objects.
- g) On Line Vibration Monitoring Systems should be in service.
- h) Vibration Isolation System (Crusher Spring Deck, Dampers) should be clean.
- i) Scoop Healthiness shall be checked.
- j) Oil Cooling System (Water flow) should be in working condition.
- k) Communication facilities like PA system /Mobiles are available.

Annual Overhauling of Crushers should be done wherein dynamic balancing of rotor assembly shall be carried out in situ.

Vibration Isolation System (Crusher Spring Deck) must be checked in house every three months and jointly checked with OEM/Expert once a year. Standard Checklist for the same is attached as Appendix-IV.

2.5.1 Stacker cum Reclaimer

Stacker Reclaimers are provided for stacking the crushed coal and to reclaim the coal, as and when required.

The following points need to be checked before returning Stacker Reclaimer into service:

- a. The track is clear of any obstructions.
- b. There is no oil leakage from Gear boxes, Fluid couplings & Hydraulic system. The oil levels are correct.
- c. The availability of power supply is to be checked.
- d. All operations e.g. travel, slewing, luffing etc. are available.
- e. Brakes are clean and functioning correctly.
- f. CRD cable is free from damage and any obstruction and is tensioned correctly.
- g. Auto greasing system is available and working.
- h. Sufficient Lighting is available on various platforms and approaches.
- i. Yard conveyor can be stopped from operator's cabin.
- j. Slew Drive Motor current readings shall be monitored during operation for detection of any unbalance load.
- k. Working of anemometer to be ensured.
- l. Rail Clamp should be in service.
- m. Thruster Brakes should be in service.
- n. All protection and interlock should be in service.
- o. All stations having Two Machines in same rail track should ensure the healthiness of Anti Collision Protection and its interlock.

Health Assessment of Stacker Reclaimer
shall be carried jointly with OEM/OES/Reputed

Vendor/Expert primarily to check following major points:

- a. Slew Bearing Healthiness (Half Yearly)
- b. Healthiness of Centralized Auto Lubrication Systems (Quarterly)
- c. Structure Health assessment (Yearly)
- d. Counterweight balancing if any structure work carried out on machine.

Rail Alignment /Rail Level Checking with respect to slew bearing. (Yearly)

Periodical checking of welding machine (installed in Stacker Reclaimer)
earthlings should be done quarterly to prevent slew bearing damage.

Trippers

Trippers are provided on coal bunkers to receive coal from the conveyors and feed it to the bunkers. The tripper is mounted on rails and feeds coal to the bunkers for use on all the units.

The following checks are to be carried-out before taking the tripper into service:

- a. The availability of supply.
- b. Tripper is in position on specified bunker.
- c. The brakes are clean and functioning correctly.
- d. The tripper discharge chute is free from jamming/blockage.
- e. The Festoon cable is free from any obstruction and is tensioned correctly.
- f. Conveyor belt tripping protection from tripper over speed is healthy
- g. There is no leakage from gear boxes, fluid couplings & hydraulic system. The oil levels are OK.
- h. Correct size of grill over the bunkers and sealing belt over the grating has been provided.

-
- i. Over speed protection should be in service.
 - j. EPB for tripping the tripper and conveyor should be provided in travelling tripper's operator console.
 - k. Rail Clamp and Thruster Brakes are healthy.
 - l. Coal Leveler for Rated flow should be in place.
 - m. Tripper Position Switches should be working.
 - n. Over Travel Limit Switch should be in place. Respective protection and interlock should be in service.

Wagon Tippers

Wagon Tippers receive Box type wagons and topple them one by one to transfer coal to the underground hopper which feeds apron feeders and conveyors running underneath the hopper.

The availability of Wagon Tippers is to be checked in the beginning of each shift in which unloading of wagons is to be carried out.

The following points need to be checked before taking any wagon tippler into service:

- a. There is no oil leakage from Gear boxes or Fluid couplings and the oil levels are normal.
- b. The Apron feeders, dribble conveyors are available.
- c. The receiving and discharge chutes of the Apron feeders, dribble conveyors are free from any blockage.
- d. There is no accumulation of coal below dribble conveyor. Any accumulation of coal below dribble conveyor, Apron feeder must be avoided to prevent chain and pans failure.
- e. Side Arm Chargers are healthy.
- f. Walkie-Talkies are available with operating staff and Wagon Tippler operator for effective communication.
- g. Additional protection and interlocks may be incorporated in the wagon tippler system for safe wagon top coal sampling system.
- h. For enhancing the safety of operation, additional protection & interlocks, may be incorporated at In-haul & Out-haul side manual intervention.
- i. Hooters/buzzers should be provided for SAC movement and wagon tipping operation.
- j. The movement area of Side arm charger should be barricaded.

Special care to be taken for hydraulically operated Wagon Tippler system.

1. The power pack rooms of WT and SAC are to be sealed from dust ingress and sufficient air circulation preferably pressurized air circulation to be ensured.
2. Oil Level
 - a) Oil Level should not fall below the range of sight glass of oil reservoir. This should be checked daily.
 - b) Float switch / Level switch should be checked periodically for proper functioning.

3. Oil Temperature

Thermostat should be checked periodically for proper functioning by reducing the setting temperature to atmospheric temperature.

4. Oil Filtration

- a) Check the condition of the filters daily. If the clogging indicator is in red zone, it indicates that the filter is clogged and needs replacement. One no of filter cartridges in the original packaging should be kept in stores readily all time.
- b) Operation of online filtration system is to be ensured to maintain oil quality within NAS Value 7.
- c) Periodically open the drain valve gradually and drain off any water that might have separated from oil.
- d) Change oil every 2000 operating hours or at least once a year or as indicated by laboratory tests. Clean the reservoir before refilling.
- e) In case of frequent clogging of filter either centrifuge the oil or replace the oil. Addition of oil must be through an oil filter.

5. Air Breather

- a) Never leave the air breather open.
- b) As a general rule, carry out maintenance of breather at the same time as oil filter.
- c) Pressurized oil breather system is preferred for prevention of ingress of dust / coal particles.

6. Oil Cooler: The cooler tubes should be cleaned quarterly or as required.

7. Hydraulic Lines & Fittings: Daily visual checks to be done for wear, damage and oil leakages. Tighten, repair or replace the components as required. Also check for collapsed suction compensators.

8. Hydraulic motor should never be run in dry condition. Refer to O&M manual of wagon tippers.

Important Precautions to be noted in O&M of Wagon Tippers:

- Wagon tippler should never be run in maintenance mode with wagon.
- Do not tip with the clamps in fully raised condition in Maintenance mode.
- Do not restart the tippler which is tripped by over travel limit switch without investigation the cause, specially, check the condition of rotary cam Limit switch.
- Wagon tippler not to be rotated in empty condition (i.e. without wagon), and before removing any braking system or decoupling, maintenance manual to be strictly followed as tippler will go towards it's zero position (which is not horizontal in Simhadri wagon tippers).

It is recommended that Wagon Tippler Health Assessment & Counterweight balancing should be done on yearly basis, whereas hydraulic system health assessment should be done quarterly by OEM/OES/Expert/Reputed vendors.

As per CC OS IOM “OS/BoP/IOM/2021-16” following must be ensured:

- a. Start/Stop push button to be provided in vicinity of dribble conveyor with Start operation in inching mode
- b. Provision of DC lights, in case of power failure, should be available in Wagon Tippler Complex underground/tunnel
- c. Control of all equipment located at wagon tippler complex viz. wagon tippler, apron feeder, dribble conveyors etc. are to be provided from wagon tippler control room.

All requisite provisions mentioned in OGN-OPS-SYS-037 Titled “Common Standard Operating Procedures for Wagon Tippler Operations” to be complied.

Coal Samplers

In the coal handling plant one coal sampler is provided before the crusher house to take a sample of coal-as-received, with a second coal sampler before the bunker conveyors to take sample of coal.

The following points need to be checked before taking coal samplers into service.

- a. The crushers are clear of any jamming.
- b. The inspection doors of transfer chutes are closed.
- c. There is no oil leakage from gear boxes, fluid coupling or the hydraulic system. The oil levels are correct.
- d. The sample collectors are empty.
- e. Interlocks and protections given in 2.1.20.

Vibrating Grizzly Feeder/ Vibrating Feeders

- a. There is no leakage from gear boxes, fluid couplings etc. The oil levels are OK.
- b. Inspection doors are properly sealed.
- c. Vibrating screen is free from jamming.
- d. V-Belts are in place.
- e. Proper guards are at place.

SECTION-IB : MAINTENANCE OF COAL HANDLING PLANT

MAINTENANCE

The over-all objective of planned maintenance is to achieve **high reliability and availability** of the plant and to **ensure safety of the equipment and the facility**.

Preventive Maintenance (PM)

Preventive Maintenance is a periodical maintenance of the equipment to minimize break-downs and increase the availability.

Scheduling of Preventive Maintenance

The equipment requiring Preventive maintenance will be identified and the periodicity of the work decided based on the type of equipment, application and type of P.M. work etc. For better maintenance and effective utilization of equipment, the periodicity could be weekly, monthly, quarterly or half-yearly which also depends on manufacturer's recommendations. This can be referred to as PM-1, PM-2 and PM-3 and the activities considered in PM-1, PM-2 and PM-3 for main equipment are to be detailed as per the Maintenance formats.

Daily planning meeting is to be conducted for CHP and FC in presence of MTP representative. Apart from discussing plant status preventive maintenance schedule for the next day is to be discussed and agreed upon.

The PM will be successful only if the PM is inspected by competent staff. To ensure correct inspection responsibility centre's are to be identified apart from random inspection by HOD (Fuel Mgt.).

Overhauling and Preventive Maintenance scheduling for control panels, motors, switch-gears and other protections are to be followed.

All relays are to be calibrated periodically at least once a year and defective relays replaced.

Predictive Maintenance

The Department should carry out Condition Monitoring of equipment as per MTP. A schedule is to be prepared, so that the Condition Monitoring of all the equipment is completed within a month. Based on this, the Planning Department shall issue Work Order Cards for rectification or preventive replacement of defective equipment.

Additionally, department should possess handheld/pocketable instruments for measuring temperature and vibration of equipment as and when required.

All Operator check-lists and Engineers walk-down check-lists are to be implemented regularly and the data listed in the check-lists, to be processed by an identified Engineer in the Operation Department and accordingly predictive maintenance works to be identified and work order cards issued to the maintenance departments.

Break-down Maintenance

The following are some of the major break-downs which occur in a Coal Handling Plant.

- a) Conveyor belt snapping
- b) Pulley failures
- c) Crusher hammers missing/breakage and high vibrations of crusher.
- d) Damage to the equipment due to the presence of foreign materials in the coal.

The above break-downs can be prevented by implementing the following:

- a) In a conveyor belt, the number of joints should be kept minimum. Care should be taken during vulcanizing for proper alignment to avoid swaying of conveyor

-
- belt. The distance between the joints for long conveyor (500 meter and above) should not be less than 200 meters. In any case the distance between two joints should not be less than 50 meters.
- b) The conveyor belt to be examined daily and if any weak joint is noticed, it should be repaired immediately.
 - c) As a general practice, hot vulcanizing method of joining the belt is to be adopted. In emergency conditions, cold vulcanizing may be adopted. Joining of belt by using clamps should be completely avoided.
 - d) Welding works of pulleys are to be Dye Penetrant tested once every two years, and corrective action is to be taken.
 - e) Practices of Belt mapping, idler mapping, Pulley and chute mapping to be adopted as per CC OS guidelines as illustrated in Appendix-V to Appendix-VIII. Critical area of structures to be D.P. Tested once a year as per guide lines issued by Corporate Centre.
 - f) Crusher vibration to be continuously monitored through vibration indicators mounted in CHP control room. Vibration level of Crusher House structure be monitored and deflection of key members be recorded periodically.
 - g) A history of break-downs is to be maintained.
 - h) Belt cleaners should be made available and maintained in service continuously.
 - i) No protections should be by-passed.
 - j) Magnetic detector and In-line Magnetic Separator/Suspended magnet availability are to be ensured.
 - k) While greasing of the pulley bearing should be thoroughly cleaned by tarpin oil and bearing should be checked for any damage before putting fresh recommended grease.
 - l) Chutes are to be regularly inspected so that loose liners are identified to prevent damage of the belt/mills due to dislodging of the liner plate.

Availability of resources, tools and plants

In achieving effectiveness and efficiency in maintenance management systems, the availability of resources, tools and plants (like Hydraulic Jacks, Laser alignment kits, Pneumatic Torque wrench, Plasma cutting machine, Welding Machines, Pipe cutter, Bearing Pullers etc) play an important factor. Besides Spare Parts, availability of trained manpower and tested tools and plants should also be planned. Frequently maintenance work is stopped due to the non-availability of skilled manpower or certain special tools. These aspects should be properly planned and included in contract SOWs if required through contract.

Maintenance Facilities

The following maintenance facilities assist in reducing down-time and returning equipment to service promptly.

- a. Permanently placed winches at Crusher House, Near Bunker areas and high Transfer Points.
- b. Chain pulley blocks in all take-up platforms to hold the take-up in position to relieve belt tension during vulcanising, pulley lagging etc.
- c. Providing a lift in the Crusher House & high rise Transfer Points where frequent inspection is required.

-
- d. Providing adequate number of chain pulley blocks at all Transfer Points.
 - e. Providing electric winches at all strategic locations for belt laying.
 - f. Providing a 24 V D.C. supply at all Transfer Points and Crusher House for lighting inside the chutes, while inspection or replacement of the liners inside the chute.
 - g. Providing access platforms for hoist maintenance.
 - h. Preventive maintenance and regular inspection and load test certification of Hoists & winches should be done.
 - i. Providing access to Transfer Points to ensure easy transportation of materials.
 - j. Regular cleaning of pulleys, coal flow chutes & gallery deck plate
 - k. Providing an access platform for operation and maintenance of Flap Gates.
 - l. Providing additional manholes in chutes for inspection and repair works.
 - m. Providing compressed-air and service-water system for cleaning.
 - n. Making available maintenance booths with welding Transformer, oxygen and dissolved acetylene cylinders at strategic locations/Transfer points.
 - o. Provision of Electrical supplies for welding machines or vulcanizing equipment.
 - p. Provision of stands for rolls of new conveyor belting to enable new belts to be unrolled.
 - q. Provision of fixed rollers on underside of conveyor steel work to guide new belt onto conveyors and old belts away from conveyor.
 - r. Developing and maintaining a small site lubricant store in CHP area.
 - s. Maintaining a minimum stock of fast wear steel plates/components at CHP workshop.

Maintenance of Belt Conveyors

For efficient and reliable operation of belt conveyors, the following criteria should be followed:

- a) Carry out a systematic and careful inspection of individual parts.
- b) Undertake regular and proper lubrication of mechanisms.
- c) Essential major and minor repairs should be carried-out promptly.
- d) The alignment of pulleys, idlers etc., should be thoroughly checked.
- e) Idlers should be used throughout the length of conveyor. Care must be taken to see that all idlers and rollers rotate freely and are fixed rigidly to their supports.
- f) Periodic checking should be carried-out to ensure that:
 - i. The belt runs centrally.
 - ii. The belt is properly tensioned.
 - iii. The belt joints are strong.
- g) Worn skirt rubber to be replaced immediately to prevent spillage. New skirt rubbers must not be so large so that they block coal path on belt.
- h) The correct counter-weights should be maintained on all conveyors and adjusted in accordance with design data. Also ensure freedom of their

movement.

- i) Maintain correct oil level in Fluid Couplings to avoid slippage, over loading, and also to maintain proper torque. This is to be checked and corrected, during PM.
- j) Take up area is to be kept clear all the time.

MODEL JOB PLANNING

- a. Everyday CHP MM head to decide jobs to be taken up next day, after consulting area executives.
- b. Jobs to be planned are
 - PM Jobs (As per schedule /sequence/priority),
 - PM path/same path defect notification jobs,
 - PM path/same path walk down checklist jobs,
 - PM path/same path belt mapping/idler mapping jobs etc.
 - Safety Observations noted during daily surveillance.
 - All jobs to be clubbed as far as possible.
- c. CHP MM head to discuss job plan with CHP Operation head, before lunch hrs, and finalize.
- d. In MTP & CHP meeting after lunch, detail jobs are discussed in presence of MTP, Shift in-charge and plans are firmed up. Support required from Electrical, C&I, Housekeeping also to be discussed.
- e. Along with job planning, planning for resource mobilization also to be done (i.e store issue, heavy vehicle requisition, availability of hoists, winches, shifting of material, scrap MRN etc).
- f. CHP MM to have a formal meeting with all concerned agency supervisors during evening regarding job planning done and instruct them to mobilize accordingly. This will help in timely mobilization in next day.
- g. Copy of planning sheet should be available at CHP control room. PTW issue may be done in night shift end or morning shift beginning.
- h. Next day, at morning, there should be a standing 10 to 15 min meeting at CHP control room with presence of all concerned executives including Electrical, C&I, MTP. Shift charge Engineer to brief regarding site condition, problems, PTW issue etc. Sometime, as per site condition, there may be changes in planning, which is to be discussed and re-planned.
- i. PTWs are issued, resources mobilized and then planning cycle for next day starts.

MAINTENANCE OF EQUIPMENT HISTORY:

History of following equipment to be maintained:

- Crusher
- Paddle feeder

- Wagon tippler
- Apron feeder
- Vibro Feeder
- Stacker reclaimers
- Conveyor belt
- Conveyor drive
- Travelling Tripper
- BF/RBF/MBF

SAFETY:

Equipment Safety

The following points need to be followed for the safe operation of equipment.

- a. Start equipment with proper clearance from the local operator after a thorough inspection.
- b. Stop the equipment immediately if any abnormality is found.
- c. Ensure the safety-guards on all rotating equipment in position.
- d. Bypassing of any protection should be in accordance with the Operation Directive OD/OPS/SYST/005 Titled "Procedure for Regulating the Bypassing of permissive and protection".
- e. The Shift engineer should be informed about the equipment under permit and the PTW should be displayed in the Control Room.
- f. Operate all the equipment in accordance with the manufacturer's instructions and any special instruction from the shift engineer
- g. PTW must be issued with JSA (Job Safety Analysis) for every job.
- h. No Notification/WOC to be closed without PTW/LWC. PTW detail should be entered in the register.
- i. The lockout box or LOTO should be used while issuing any PTW.
- j. PTWs are to be issued through SAP with isolation and normalization slips.
- k. EPB must be close to the system, easy to access, and free of debris and obstructions
- l. To give PTW on water systems if manual Valve is closed, it must also be locked with chain and Lock or LOTO valve handle locking system, to avoid accidental opening.

The following points need to be observed for safe maintenance of the equipment:

- a) Maintenance should be undertaken only when the permit is issued in SAP
- b) Only correct tools should be used together with tackles and rigging tools required for the particular type of work and equipment.
- c) Oils and Gases are to be handled with care.
- d) Hot permit is to be taken for cutting/welding works. Fire prevention is to be provided (fire extinguisher, water hoses). Fire blanket can be used to cover conveyor belts or any other component which may cause fire during work. No welding/gas cutting work should be carried out at locations of chutes/hoppers

having synthetic liners/rubber backed synthetic liners fixed from inside. If any welding job has to be carried out in such fire hazardous zones, the liners should first be dismantled prior to the welding/cutting job. And after completion of any welding/cutting activity proper quenching of hot surfaces and combustible materials like coal, conveyor belt, pulley lagging etc. should be ensured. Thereafter proper inspection is to be carried out and observations recorded by Engineer-in-charge.

- e) Follow the instructions, detailed in the work specification to ensure safety during work.
- f) Use the correct scaffolding while working at heights.
- g) Remove all tools, tackles & unwanted material after completion of maintenance.
- h) Preferably Agency carrying out Welding/Cutting must keep a water type fire extinguisher near the work it is carrying out to handle any fire earliest.

All tools & tackles are to be tested and healthiness to be certified by authorized agency.

Human safety

Human safety is paramount factor in the industry premises. It plays a vital role in production & productivity and utmost care must be taken to implement all necessary safety requirements.

The following are the general safety guidelines that apply in the Coal Handling Plant.

- i. Housekeeping is most important parameter to indicate healthiness of any CHP. It should be monitored and maintained by proper deployment of manpower and attending defects leading to spillages earliest.
- ii. Lighting /PA systems are a must to ensure safety.
- iii. Daily Pep Talk, Toolbox Talk and Safety oath must be conducted among all workers/working staff by CHP executives.
- iv. There should be safety supervisor for each contracts.
- v. Daily departmental cross functional safety surveillance should be done. (Appendix-III)
- vi. Use only cross-overs bridges to move across the conveyors.
- vii. Keep away from the rotating equipment e.g. Couplings, fans, etc. and provide protective guards around them as much as possible.
- viii. Wear protective safety equipment and avoid wearing of loose clothing. All electrical isolations to be done wearing Arc Flash Suit, Use of Fuse Pullers only (No Pliers), No use of Contact line testers.
- ix. Do not open the Manholes and Inspection doors of the Crusher/Chutes/Coal samplers whilst in operation.
- x. Use only approved and tested tools for maintenance.

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- xi. All stone-pickers should operate from a proper platform and should use full body safety harness They must have dedicated EPBs to trip conveyor.
 - xii. Walk-ways should be free from obstruction and adequately lit throughout the plant.
 - xiii. Stair-case steps and hand-rails should be properly arranged with proper toe-guards.
 - xiv. Floor openings are to be properly covered, side openings for doors, and hoists should have hand-rails.
 - xv. Sufficient walking space should be provided at the drive area of equipment.
 - xvi. All electrical installations and panels are to be kept closed.
 - xvii. Welding connections are to be taken from the sockets provided.
 - xviii. Wherever swing doors are not provided at bottom to Track Hopper, a minimum bed of coal should be maintained in hopper to avoid coal spillage over the personnel working in the Track Hopper area.
 - xix. Ensure fencing of all the Gravity take off areas in ground/internal to floors.
 - xx. The area underneath the stone picking platform should be fenced to prevent entry.
 - xxi. Periodic checking of hoists and lifts should be carried out.
 - xxii. Ensure all guards are secure in place.
 - xxiii. Ensure healthiness of all pull cord switches along with ropes.
 - xxiv. Ensure healthiness of all EPBs. Ensure installation of EPBs as close as possible to drives.
 - xxv. Ensure healthiness of hooters.
 - xxvi. The available conveyor 'start' alarms should be used and tasted regularly in Event logs.
 - xxvii. All safety surveillance reports should be reviewed by CHP I/c and FM I/c during daily CHP planning meeting.
 - xxviii. Safety surveillance exception points to be reviewed by GM (O&M) and HOP.
 - xxix. In Case of Patch Repair belt should be clamped from both sides
 - xxx. If the Belt is in choked/overloaded condition, in no case person should step on belt / Deck sheet/Seal plate even though it may be electrically isolated because in spite of belt being isolated, it may be in tension and as soon as choking gets cleared the tension will get released and belt can move .In Case of person working on the belt like belt patching, there should not be a cross PTW on Gear Box / Coupling of Conveyor, because in case Holdback of Gear box is
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disturbed or coupling is opened during the activity, the belt will roll back at very high speed.

FIRE PROTECTION

Coal Handling Plants are prone to fire accidents due to fine coal dust accumulation and chute jamming.

The following facilities are to be installed and maintained to detect fire in Coal Handling Plant:

- a) Linear-heat sensing cable all along the conveyor with an associated alarm system in Control Room.
- b) Infra-red fire detection cameras at all strategic locations especially on reclaim conveyors, for detection of moving fires.
- c) Medium velocity sprinkler system along with quartz bulb type detection-cum-actuation system for automatic fire detection and protection.
- d) Fire hydrants at Coal stock yard, Track hopper area and Transfer Points.
- e) Fire alarm system to be installed throughout the coal handling plant.

The following facilities are to be installed for fire protection :

- a) Fire hydrant lines at all strategic locations with necessary hoses and Couplings. Minimum hydrant water pressure at all locations must be 3.5 Kg/cm^2 except booster pump fed locations. At booster fed locations it should be around 2 Ksc, without booster pump in service. Routine monitoring of hydrant/sprinkler water pressure at 4/5 strategic locations particularly Highest and farthest location must be done. Provision of permanent pressure gauges to be made and appropriate log sheets be maintained. Routine checking of Booster pump auto start must be taken and seen that water pressure rises up to at least 3.5 KSC.
- b) Portable fire extinguishers for all types of fire.

All personnel should be trained in the use of fire extinguishers and fire fighting and should be instructed not to light unauthorized fires.

Emergency instructions should be prepared and displayed in the Control Room.

A register is to be maintained separately for the Fire Hydrant and Emulsifier systems. Any isolation/permit/defect/communication is to be entered in the register.

The whole Coal Handling Plant should be inspected by the Safety and Fire Fighting group on a fortnightly basis. Corrective action should be initiated for any defects found and reasons for prolonged isolation/permit investigated.

Fire hydrant and sprinkler pressures should be monitored, daily.

Various Fire Accident Committee recommendations are to be studied and implemented to avoid recurrence of such fires.

Good housekeeping helps in reducing fire hazards. The following practices should be established in order to maintain good housekeeping in the coal handling plant.

- a. Removal of all metal and belt scrap, debris and coal from all Transfer Points and galleries.
- b. Clean all cable-trays, bracings and not accessible points regularly.
- c. Providing packing for all manholes to prevent leakages.
- d. Water should be sprinkled before sweeping and water washing of galleries should be carried out periodically.
- e. All spillage or accumulation of coal should be removed regularly.
- f. All DE and DS system to be kept healthy and in running condition

Requisite provisions for fire protection and detection for CHP, referred in OGN-OPS-SYST-002 has to be complied.

In view of the use of Biomass Pellets all necessary care must be taken for fire hazards as per OGN-OPS-SYST-049.

PERFORMANCE MONITORING

The performance of the plant Operation and Maintenance are to be monitored to improve the efficiency and increase the availability and reliability of the plant. The performance is monitored Daily, Weekly, and Monthly as mentioned here under.

Daily performance monitoring: Daily performance of plant utilization, reasons for outages, equipment availability and fire hydrant pressures at the farthest locations are to be monitored.

Weekly performance monitoring: All the work order cards raised and completed in the week are to be monitored and analyzed to identify any delay in completion.

Monthly performance monitoring: The overall performance of the plant for the month, is monitored in respect of the following:

- a) Availability of equipment.
- b) The Crusher hammer life.
- c) Crushing hours of each crusher during the month.
- d) Spares consumption e.g. belt and hammers, fluid coupling, gear boxes etc.
- e) Belt jointing, pulley laggings done.
- f) Major works attended during the month.
- g) Utilization factor of CHP's (Stage wise).
- h) Availability of auxiliary equipment.
- i) Any spares reclaimed.
- j) Availability of protection.
- k) Reclaiming Capability test and yard coal distribution monitoring as per OS Guidelines.

An Action Plan of various works are to be made as per the format decided by station and shall be monitored regularly.

Reporting System

- A Daily report (Refer Appendix-II) is to be prepared at station level and discussed daily in DPM.
- A Monthly report (ORT sheet "CHP.xls") is to be prepared to be sent to HOD (BoP), Corporate OS.
- A yearly Highlight report is also to be prepared and circulated through Station EEMG.

Benchmarking of Performance Parameters:

Benchmarking will be as per marks of RIM vide OIN "COS-ISO-00-OIN/OPS/SYST/075". CHP RIM marks of each month shall be evaluated in association with ROS during pre ORT site visit and exceptions to be discussed in ORT.

Capacity Test of Coal Handling Plant shall be carried out as per OGN-OPS-PERF-009 (Latest Revision) Titled Performance and Capacity Test of Coal Handling Plant.

9.0 MODIFICATION RECORDING

All modifications carried out on various equipment/system should be as per OD-Gen-006 Titled System for Plant Modifications. Requirement of such modifications, nature of problems faced and effect of modifications should be mentioned. Before carrying out any modifications specific approval should be obtained.

SEASONAL PREPARATIONS

Before the ensuing summer and rainy season specific work program be prepared and executed to ensure smooth operation of the plant.

Summer Preparation: following to be taken care of

- a. Availability and operation of Dust suppression systems.
- b. Availability of Yard Spray system with self-rotating sprinkler nozzles.
- c. No dry grass near CHP systems.
- d. Availability of Fire protection systems
- e. Regular cleaning of all areas to ensure no coal dust accumulation.
- f. Yard management as per clause 12 below.

Monsoon Preparation: Detailed guidelines for monsoon preparation are given in Section-IV from Clause No. 13 of this document.

Winters: There is a tendency to burn coal for heating in CHP. It must be ensured that this is avoided to prevent CO asphyxiation or fire risks to Conveyors.

SECTION-II: COAL STOCKYARD MANAGEMENT

COAL STOCKYARD MANAGEMENT

This section of Guidance Note specifies the requirements for Management of Coal Stock yard effectively, economically and safely. For storage of coal this document shall be read in conjunction with the various guidelines issued by CC-FM from time to time.

Coal stacking to be done in such a way to prevent cyclic deterioration in coal quality, to prevent/minimize spontaneous ignition of coal, to prevent sliding of coal particularly during monsoon and to help in ease of reclamation of coal whenever required.

Coal piles should be trapezoidal in shape with their base having sufficient clearance from rails and drains. Height of pile should not exceed the maximum designed value (Generally 10 m in NTPC stations).

Each pile should have covered surface drains on both sides all along the length. Inner wall of drain (towards pile) should have holes and should be back-filled with graded gravel and stone chips.

In case of frequent auto ignition incidences in coal stockyard temperature of piles at different locations to be measured at various depths so as to know the trend and to decide as to which coal is to be reclaimed first. Temperature of coal stockpile can also be monitored by installing swiveling infrared camera on high mast located strategically at corner/corners of stockpiles.

During stacking coal in the piles should be spread out in horizontal layers of 1-1.5 m depth and not in conical heaps and compacted for further stacking.

Freshly mined coal should be stacked and old coal should be used for reclaiming. An important aspect in the good management of coal stack pile is the First in First out (FIFO) principle.

Coal air interaction is the most important phenomenon for auto-oxidation and spontaneous combustion. Dry coal-wet air is the most critical combination and moist coal-dry air is the least critical combination for spontaneous combustion. In fact the latter combination takes away heat from coal and lowers its temperature.

March-June is the critical period for spontaneous fires and close monitoring should be done during this period by regular inspection of measuring temperature of coal stock pile.

In case freshly mined coal has to be stacked over old coal, latter's temperature should first be measured. If the temp. is more than 50°C coal should be drenched with sufficient water.

Lumps and fines should be stacked together and compacted.

Before stacking, the coal should be visibly wet so that condensation of vapors in

~~the air never takes place. Rather evaporation of moisture from coal to~~
atmosphere should

keep coal stockyard cool.

Each stockpile should be segregated into various segments and records of period of stacking, quality of coal (GCV) should be maintained. Coal of these segments should be used based upon requirement and period of storage (i.e. on first IN, first OUT basis).

It has been observed that while stacking bigger particles of coal get segregated and roll on bottom sides. More interaction of such coal with air causes self-ignition. After stacking, if finer coal particles are placed over this segregated bigger coal particles, and compacted, this shall reduce the chances of fires.

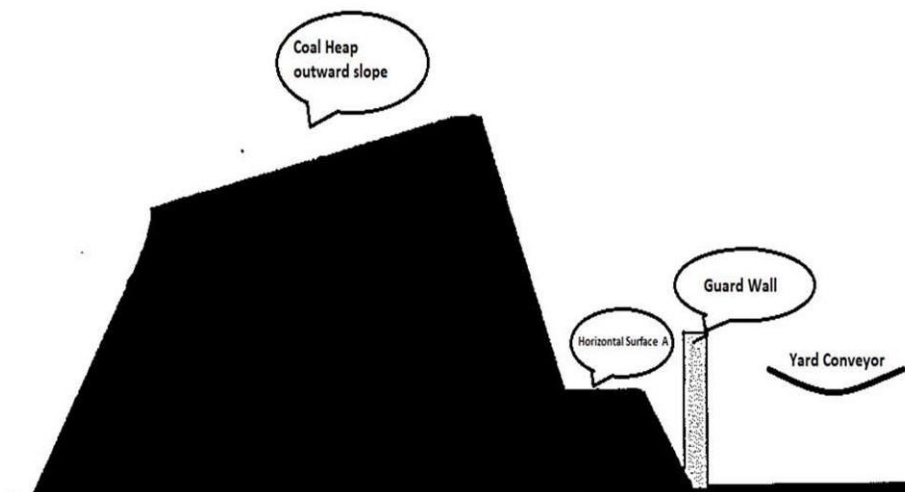
Stacking of coal is to be done layer by layer (each of 1 to 1.5M height) and compaction of each layer by dozer is to be done separately to lessen the chances of sliding down during heavy rains and of spontaneous ignition of coal.

Coal stockpiles are to be kept covered by polythene sheets during rainy season. (picture attached). All stations which are facing problem with wet coal or are expected to face such problem are required to cover their coal stockpiles (at least one or two stockpiles) with polythene sheets. This will prevent stacked coal to become wet /sticky. The sheets are to be kept in place against the wind by keeping empty cement bags filled up with coal (as shown in the picture) on them.

For yard covering, stations are to procure LDPE sheet well in advance, so that stock of these LDPE sheets are available before June every year as per Approved Technical Specification of LDPE sheet.

Top surface of stacked coal to be sloped outward for ease of water draining. There should not be any concave cavity over the top surface in order to avoid formation of water pool over there. While compacting coal layers with dozer the sides don't get compacted. Therefore, 3-4 m wide coal from inner wall of stockpile is to be removed with stacker reclaimer as shown in the sketch and compacting the horizontal surface 'A' so formed with dozer.

In a few stations coal feeding gets hampered during rainy season due to sliding of coal of yard to the conveyor and surrounding areas. In NTPC's new stations guard walls (with holes to allow flow of water to drains) are provided in the yard to prevent coal sliding. Stations, where guard walls are not provided, may study the feasibility of providing guard wall in the conveyor side (as shown in the picture). If found feasible, the same may be constructed.



Coal should be piled in such a manner that air can circulate freely to dissipate heat. For crushed coal the pile should be packed thoroughly to ensure the complete absence of air.

Coal piles should be trapezoidal in shape with their base having sufficient clearance from rails and drains. Height of pile should not exceed the maximum designed value.

Piles should be regularly inspected for high temperature ignition of coal. Each pile should have fire water hydrant valve all around the periphery. Temperature of pile at different locations can be measured at various depths so as to know the trend and to decide as to which coal is to be reclaimed first. Hot spot detection by thermography can also be done to detect hot spots and take requisite action to prevent fire/GCV loss.

The toe and the edge of the pile should be compacted to avoid spontaneous ignition.

It is desirable to have wind barrier/wind fence to prevent fugitive dust emission. Also core temperature of the pile can be monitored by measuring with a suitable thermo-vision camera.

Illustration for Covering of Stock Pile during Monsoon.



Recommendations for practicing FIFO:

- a. Maintaining coal stacking data clearly mentioning the date of stacking and the zone in which the mentioned coal is stacked.
- b. To the maximum possible extent the stacking shall be preferably done longitudinally in same zone during same day with adequate compaction after every 1.5-2.0 meter height.
- c. To minimize the frequent long travel movement of stacker reclaimer the stacking operation is recommended to be done at diagonally opposite zones/sub zones of particular stacker reclaimer for different types/grade of coal.
- d. During Reclaiming, the zone with oldest stock shall be preferred for reclaiming.
- e. It is recommended to use the coal stored in yard within three months of stacking.
- f. Coal shall be reclaimed via a series of advancing terrace cuts. After the top bench has been partially reclaimed to a set distance so that the next bench is commenced. The subsequent intermittent benches must end prior to the bench above to prevent material slump and collapse of the digging wall.

In order to preserve coal quality at stockyard, compaction level of coal at stockyard to be measured with the use of Dynamic Coal Penetrometer as per CC-FM guidelines Ref: CCFM/CGM/Coal Data/02 Dated. 04.02.2020, compaction index to be maintained approx. 2.5 cm -12.0 cm.

Stockyard to be divided into different zones for age analysis and GCV mapping/profiling.

Monthly GCV mapping to be done with the help of Chemistry and record to be maintained.

Age analysis of coal at stockyard to be done and record to be maintained. Zone wise age of stock with GCV to be displayed at CHP Control Room as per CC-FM guideline 01/CC/FM/CGM Dated. 04.01.2020

In order to monitor coal compaction, dozer utilization and rationalization of diesel consumption, GPS device can be installed on all the Dozers of CHP as per CC-FM Guidelines CCFM/CGM/Dozers/01 Dated. 22.01.2020



Illustration of side guard wall in coal yard to prevent coal sliding.

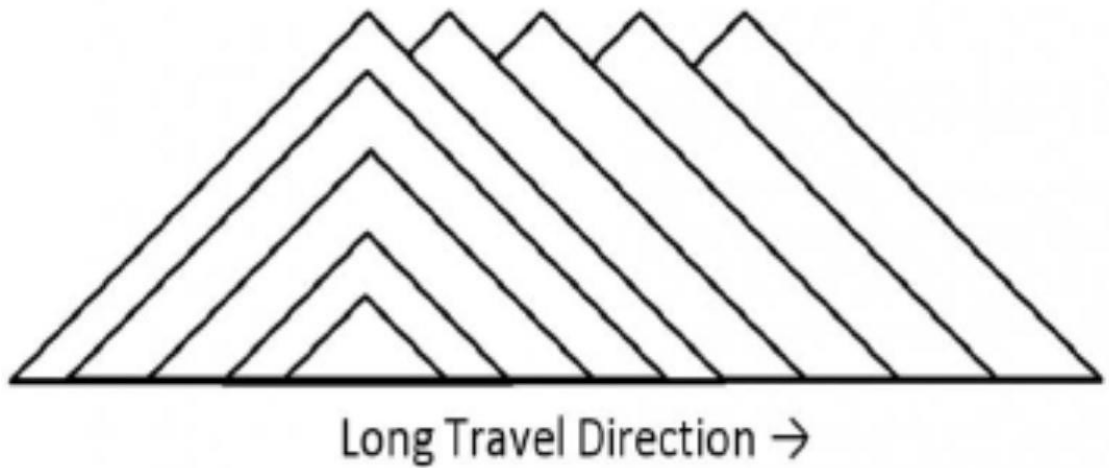


Illustration of Longitudinal Stacking



Illustration of Reclaiming Method for FIFO

Coal yard parameters(Dated:)									
Yard Stage-2					Yard Stage-1				
ZONE C1		ZONE D1		Stacker Reclaimer-1	ZONE A1		ZONE B1		Stacker Reclaimer-2
Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)		Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)	
7000	3775	4000	3753		2000	4174	20000	3474	
Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)		Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)	
1	0	4	0		0	0	1	0	
Pl(cm)		Pl(cm)			Pl(cm)		Pl(cm)		
3.50		3.40			2.85		2.40		
ZONE C2		ZONE D2			ZONE A2		ZONE B2		
Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)		Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)	
2000	3810	1000	3719		5000	1	20000	3154	
Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)		Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)	
1	0	1	0		1	0	2-3	0	
Pl(cm)		Pl(cm)			Pl(cm)		Pl(cm)		
3.90		2.80			2.81		2.90		
ZONE C3		ZONE D3		ZONE A3		ZONE B3			
Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)		
0	1	0	1	5000	3645	15000	3637		
Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)		
0	0	0	0	1	0	2-3	0		
Pl(cm)		Pl(cm)		Pl(cm)		Pl(cm)			
0.00		0.00		3.70		3.20			
Zone C Total		Zone D total		Zone A Total		Zone B Total			
Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)	Stock(MT)	GCV(Kcal/Kg)		
9000	3783	5000	3746	12000	2215	55000	3402		
Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)	Age(Mth)	CO (ppm)		
1	0.00	1-2	0.00	1	0.0	1-3	0.0		
Pl(cm)		Pl(cm)		Pl(cm)		Pl(cm)			
2.47		2.07		3.12		2.83			
AVG AGE OF STOCK YARD				3.0		Months			
TOTAL COAL STOCK (Capacity: 600000 MT)				81000		MT			
GCV:KCAL/KG				3290		Kcal/Kg			

Note: all figures are illustrative only

Illustration of Reclaiming Method for FIFO

SECTION-III: BLENDING OF COAL

GUIDELINES FOR PROPER BLENDING OF COAL

Imported coal is being received at many of the coal stations of NTPC for blending with domestic coal. In order to achieve consistent combustion characteristics and smooth out the extremes for stable operation of boiler, it is desirable that coal being fired is having uniform properties. A well homogenized coal is expected to increase flame stability, decrease furnace & steam side temperature fluctuations and reduce specific coal consumption. During the process of blending, the coals are to be intimately mixed together in a specific ratio to obtain a product with well-defined specific properties and of uniform quality with GCV going maximum up to Best Coal design of station. There have been mill explosions, clinkering in some of the stations, one of the reasons for which was found to be improper blending of coals.

For making a homogenous mixture of imported and domestic coal the following blending guidelines are to be followed:

Imported coal is first to be stacked in identified yard at earmarked stockpiles.

Mixing of imported coal with domestic coal in track hopper is to be avoided.

GCV of imported coal lying in different stockpiles is to be measured at regular intervals and record is to be made available in CHP Control Room.

Imported coal is to be drawn from stockyard through stacker reclaimer / underground hopper / over ground hopper and is to be mixed with domestic coal (being conveyed from track hopper/ wagon tippler) at the merging transfer point in a pre-defined ratio. Further homogenization for a more uniform product takes place in upstream transfer points.

Under no circumstances Import coal should be directly sent to bunker.

13.6 In case of high GCV coal received from domestic sources such as (ECL, NEC etc where GCV

is more than 4500), no imported coal blending is to be done.

13.7 Belt weighers of evacuation conveyor stream from track hopper / wagon tippler and stacker cum reclaimer machines are to be kept operational with rated accuracy and good communication between control rooms, paddle feeder operator, stacker reclaimer operator is to be established for maintaining blending ratio.

Each coal station should frame 'station specific coal blending guideline' giving equipment/conveyors (shown on CHP layout), stockyards/stockpiles to be taken in service for the purpose of blending.

GCV of coal received from high GCV domestic coal sources to be analysed in express lab, so that GCV is known to CHP control room within 2 hrs.

Details of incoming coal rake should be noted by CHP to know source of coal (i.e. company/mines / grades).

- i. From Indian Railway FOIS site
- ii. From RR

iii. From Slip pasted on coal rake by knowing the source of the coal

CHP control room to keep a data base of GCV of various mines, so that by knowing the mine's name, Control room can guess about the GVC of the coal.

If there is no feeding /flow from domestic coal stream, then after 2 minutes import stream should be tripped.

CHP Control Room engineers to be sensitized of detrimental effect of rich blending on boiler like mill/boiler explosion.

SECTION-IV: MONSOON PREPARATION

GUIDELINES FOR MONSOON ACTION PLAN

All the coal stations are advised to prepare monsoon action plan well in advance and monitor the same in various forums like Daily Planning Meeting, ORT, SMC etc.

Though the monsoon action plan will be specific to individual station requirements, however suggested common nature of actions are given below and a checklist also attached in Appendix-X:

- a) All stations which have faced problem with wet coal or are expected to face such problem are advised to cover all their coal stockpiles (with LDPE sheets. This will prevent stacked coal to become wet /sticky. For yard covering it is suggested that station procure LDPE sheet well in advance.
- b) All the drains in CHP including those in stockyards are to be thoroughly cleaned
- c) Availability of all dewatering pumps/ sump pumps especially at underground locations like track hopper, wagon tipplers, tunnels etc is to be ensured.
- d) Availability of service water pumps and decantation pumps
- e) Cleaning of settling pits and sump pits to be completed
- f) Operation of all air canons (if provided) / motorized vibrators at bunkers are to be ensured.
- g) Chutes are to be thoroughly repaired with new mother plates/liner plates for smooth flow
- h) Availability of all ZSS & CBS to be ensured
- i) Belt Scrappers are to be set right/replaced for proper functioning
- j) Worn out pulley lagging is to be replaced
- k) Critical conveyors belt healthiness is to be checked and suitably replaced
- l) Cleaning of Track Hooper & coal bunkers for any jamming inside before onset of monsoon.
- m) Effective monitoring with sufficient executives manpower is to be done

-
- n) Adequate additional labors is to be deployed at all critical TP's round the clock during monsoon
 - o) Crusher's healthiness to be ensured by replacing with new hammers. Gap between hammers and cage bar assembly to be adjusted for better scrapping of coal.
 - p) Availability of poking rods at all chutes for faster clearing of chute jamming.
 - q) Stacker cum reclaimers servicing to be done for ensuring 100% availability
 - r) Availability of water hoses with nozzles at all TPs covering all chutes and 3-4 portable ones.
 - s) Additional submersible pumps to be kept ready at suitable locations with starter and discharge hoses (at least 5nos. per CHP)
 - t) Additional diesel engine driven dewatering pump (at least three nos.)
 - u) Availability of sufficient items like raincoats, torches, umbrellas, gumboots, hand lamps, floodlights etc.
 - v) Availability of heavy equipment (JCBs, Pay loader, Tipper, Tractor trolley etc) for housekeeping during monsoon
 - w) Ensuring availability of sufficient stock of sand bags near minus meter installations (at least 100 nos. per CHP) or wherever water ingress expected.
 - x) Monitoring of coal quality at track hoppers and wagon tippler unloading areas for avoiding slushy coal.
 - y) Procurement of LDPE sheet for 100% pile covering shall be done.
 - z) Provision for manpower deployment during stacking and reclaiming operation for LDPE sheet replacement and re-fixing.

SECTION-V: Review

Review Responsibility	: CHP
Review Frequency	: 2years
Review Approval	: Head of Project

Appendix – I: Walk down Checklist Illustrative Points

Following may be checked in Walk down:

- Belt Running Healthiness (sway, Abnormal Sound etc)
- Hydraulic System Healthiness
- Condition of Idlers
- Chute Leakages
- Long Travel healthiness of Paddle Feeders/Stacker Reclaimer/Trippers
- Condition of Conveyor Drives (Gear Box & FCU Oil Leakages, Abnormal Sound etc)
- Proper working of Protections
- Availability of Requisite Guards
- Skirt Rubber Healthiness
- Housekeeping Condition
- Cooling water System Healthiness
- P.A. System availability
- Illumination Condition
- Pulley lagging Healthiness
- Availability of VIS (In Crushers)
- Travel Drive Condition, Slew Drive Condition, Bucket Wheel Condition, Boom Conveyor healthiness, CRD Healthiness, auto lubrication system healthiness, anemometer availability and storm lock etc in Stacker Reclaimer
- Seal Plate and Deck Plate conditions
- Idlers Healthiness
- GTU healthiness

COAL HANDLING PLANT DAILY REPORT

- Station Generation (MU).
- Coal Rakes Received, Unloaded, Pending for Unloading, Timings of Unloading and Demurrage if any.
- Coal Receipt, Stacking, Reclaiming, Consumption and stock data (MT).
- CHP Equipment Running Hours : (Direct)
- Belt Weigher readings and Conveyor Utilization factors
- Energy consumption calculation with Specific Energy.
- Protections not available(Like PCS/ZSS/BSS/CBS)
- Fire Protection systems not available.
- Fire hydrant and service water pressures at important locations.
- Any major equipment outage.
- Status of exceptions of Belt mapping, oil analysis, conveyor defects.
- Any Major activity or shortcoming that needs attention.

SAFETY SURVEILLANCE REPORT FORMAT

DATE	AREA	SAFETY OBSERVATION	NAME OF THE PERSON	NOTIFICATION/W ORK ORDER REFERENCE	TARGET DATE

Appendix – IV: Checklist for VIS of Crusher Deck

Checklist for Checking of Vibration Isolation Systems

(Crusher Spring Deck)

By CHP Maintenance: Once in 3 to 6 months

- Spring Units (GP Type) - Stud Bolts/Nuts to be in loose condition at least by 10mm (Minimum) during operation.
- Visco Damper Bolts/ Nuts to be checked for tightness.
- Visco Dampers Sleeves and Clamps to be checked for tightness/damages.
- Spring units / Viscodampers to be cleaned by air blowing (not by water) periodically.
- Routine inspection of the spring units and Visco dampers to be done periodically for painting/ corrosion/damages, to be spray-painted/repared.
- Crusher Foundation to be made free all around the deck slab to avoid any obstruction. Gap between the deck and adjacent floors to be maintained free of coal dust.
- In case of any major loads (machine parts) to be removed from foundation, for the maintenance purpose, the stud-bolts of spring units are to HAND TIGHTENED and later to be released after replacing the loads on to the foundation.

Note: Checking the accumulation (Once in 3 Months) of foreign materials (coal dust) around the Spring Units to be carried out, to be removed / cleaned. Frequency to be decided by NTPC respective Unit, based on the accumulation/site condition.

By OEM/Expert: Once in a Year

- Visual inspection for corrosion and painting.
- Measure of spring height of VIS. If required, adjustment to be done for spring heights according to M/s NTPC schedule.
- Check for Sleeves and Clamps damage.
- Check for Visco liquid condition.

Appendix – V: Conveyor Pulley Mapping Format

Conveyor Pulley Mapping

Date:

Conveyor No.:

Type of Pulley (e.g Head Pulley /Tail Pulley /Snub Pulley)	Lagging Condition (OK/Not OK)	Bearing Temperature (Normal/High)		Noise (Normal/ Abnormal)	Plummer Block Condition (OK/Not OK)		Any additional observations Pulley foundation corroded/Take-up structure damage
		LHS	RHS		LHS	RHS	

- Above conditions to be monitored & recorded during walk down / before PM of respective conveyor
- Bearing temperature to be measured by portable / pocketable instrument / sensor
- Safe approach to be arranged for inspection of take-up pulley
- Plummer block condition Not OK suggests e.g. cracked / bolt loose / seal damage etc.
- Apart from above Pulley weld joints to be checked by DPT once in a year, and recorded as per following format.

Type of Pulley e.g. (Head Pulley / Tail Pulley / Snub Pulley)	Observations (e.g. Bend Pulley LHS side diaphragm to shell weld crack observed)

** In view of Paperless Office this format may be maintained in soft form

Appendix – VI: Belt Mapping Format

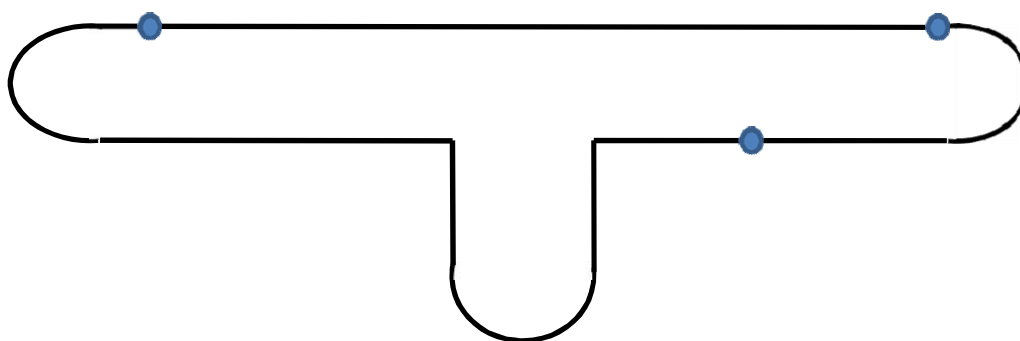
Conveyor Health Monitoring Status

Conveyor (630/1000/1250/1400/1800)

Total designated Length:

Date of Mapping:

Mapping done by:



Take-up allowance available: 0/1/2/3 Joints

SL NO.	ZONE	Span in M	Belt Condition (5-0)	Belt Make	Belt Laid on D-M-Y	Belt Span Remarks	Joint Condition ((5-0)	Joint Date D-M-Y	Jointer	Joint Solution Make	Joint Remarks

NTPC Area In-charge

Agency Representative

Notes:

- a) Rating '0' - Immediate replacement
- Rating '1' – Replacement in 15 days
- Rating '2' – Replacement in 30 days
- Rating '3' – Under observation
- Rating '4' – Condition is Good/Ok
- Rating '5' – Condition is Excellent/New

b) Span should include the stretches where different rating belts or diff make belts are available. Also it should bring out the stretches where there is reduction in thickness, width of belts.

Appendix – VII: Idler Mapping Format

[illegible]

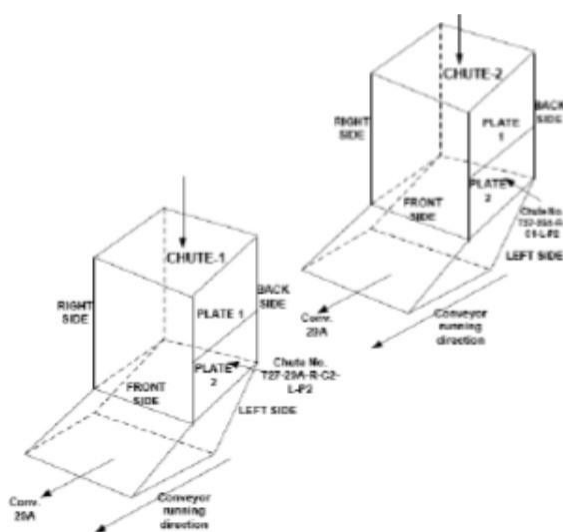
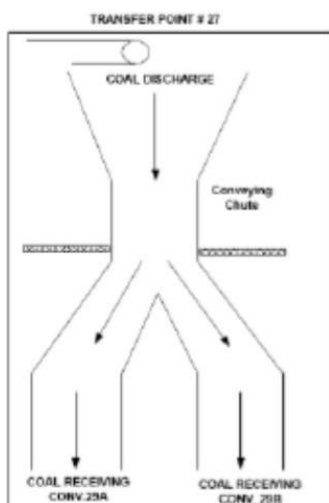
Chute Mapping

A) Chute Nomenclature

For successful mapping of chutes it is advisable to properly mark the chutes by following suggested nomenclature as illustrated below in words & figure:

Transfer Point Name, Conveyor Name, Receiving/Discharge, Chute No.(1,2,3,4), Orientation (front, back, right, left), Plate No.etc.

T27	29A	R	C4	F	P2
					
Transfer Point	Conveyor Name	(Receiving/Discharge)	Chute No	Orientation	Plate Number



B. Chute mapping format:

Date :

Conv. No :

TP. No. :

Chute Name (e.g. T27-29A-R- C4-F-P2)	Mother Plate Material (e.g. MS/ SAILHard/ Tiscral etc.)	Mother Plate Thickness	Condition of Mother Plate (eg. Leakage /No Leakage)	Liner condition (Worn-out / OK)	Liners Fixture Healthine ss (e.g. T Weld Healthy/B olts Intact etc.)	Additional Observatio ns (eg. needs painting, inspection door sealing, baffle worn out etc.)

- External Inspection to be monitored & recorded during walk down / before PM of respective conveyor
- Internal Inspection of Chutes viz liners etc., to be done only after requisite PTW has been taken.
- Requisite Cleaning and proper DC lighting to be ensured before internal inspection.
- Safe approach to be arranged for inspection of chutes.
- Liner condition Not OK suggests e.g. worn / hanging / improper zone of coal impact.
- Sealing of Inspection Door(Wherever available) in chutes shall be checked for preventing coal/ dust leakage
- Mother Plate thickness to be measured by D Meter.

C. Summary:

Signature of Area Engineer

MD Register

Date	Time	Incidence Sl. No.	MD No. Conv. No	Counter reading	Material found	Remarks

Signature of Shift Charge Engineer

Appendix – X: Monsoon Preparation Checklist

Monsoon Preparation activities in Coal Handling Plant

S. No.	Description	Status	Remarks
1.	Availability of sufficient quantity of LDPE sheets to cover all the coal stockpiles.		
2.	Availability of all sump pumps/dewatering pumps of different locations specially at underground locations e.g. track hoppers, wagon tippers, conveyor tunnels, underground hoppers etc.		
3.	Availability of additional submersible pumps(at least 4 to 5 nos) with starter and discharge hoses		
4.	All coal conveying systems including conveyors, belt joints, pulley lagging, scrappers, chute liners etc., should be healthy.		
5.	Conveyor protections like ZSS, CBS should be in service		
6.	Healthiness of crushers to be ensured by replacing with new hammers wherever required. Gap between hammers and cage bar assembly to be adjusted for better scrapping of coal		
7.	Stacker reclaimers servicing to be done for ensuring 100% availability		
8.	Coal stockpile compaction and profiling to be done to avoid coal sliding		
9.	Availability of additional manpower for deployment during monsoon.		
10.	Availability of service water pumps, decantation pumps		
11.	Availability of sufficient quantity of items like raincoats, torches, umbrellas, gumboots, poking rods, water hoses with nozzles, handlamps, sandbags etc.		
12.	Fabrication and installation of sufficient nos of inspection doors with approach platform at TPs, Crusher house, Bunker Floor etc.		
13.	Cleaning of drains, settling pits, sump pump pits, floor drains, debris chutes		
14.	Availability of JCBs, Payloader, Tractor trolleys, Tippers etc for housekeeping during monsoon.		
15.	Availability of diesel driven pumps.(At Least 04 Nos)		

List of Permit

DATE	PTW NO	CROSS PTW NO	JOB DESCRIPTION	ISSUED TO	ISSUED BY	RETURNED BY/DATE	CANCELLED BY	LOCK OUT BOX KEY NO.	REMARKS

HOT WORK PERMITS

DAT E	HW P NO	LOCATIO N/ EQUIPM ENT	DETAI LS OF WOR K	PERI OD FRO M	PERI OD TO	ISSU ED TO	ISSU ED BY	RETURN ED BY	RETU RN TIME & DATE	CANCELL ED BY	REMAR KS

Appendix – XII: Format for Performance data recording

1. COAL RECEIVED					
SOURCE	No.OF RAKES	NO.OF WAGONS		TONNAGE	
TALCHER					
IB					
CCL					
BCCL					
ECL					
SINGARENI					
TOTAL INDIAN COAL					
IMPORT					
TOTAL COAL RECIEPT					
NO OF RAKES UNLOADED AT -->	T/H-1	T/H-2	WT-1	WT-2	TOTAL
NO OF RAKES UNLOADED					
2.GENERATION	UNIT-1	UNIT-2	UNIT-3	UNIT-4	STATION(MU)
UNIT WISE GEN					
3.COAL CONSUMPTION	UNIT-1	UNIT-2	UNIT-3	UNIT-4	STATION(MT)
UNIT WISE CONS					
Specific coal					
Day's Blending	0				
4. DAY OPERATION	INDIAN		IMPORTED		TOTAL
BUNKERING FROM T/H-1+WT					
BUNKERING FROM T/H-2 (TH-1+TH-2+WT1+WT2)					
5.RECLAIMING	INDIAN		IMPORTED		TOTAL
PILE#1					
PILE#2					
PILE#3					
PILE#4					
PILE#5					
PILE#6					
PILE#7					
PILE#8					
TOTAL RECLAIMING					
TOTAL BUNKERING					
6.STACKING	INDIAN		IMPORTED		TOTAL
PILE#1					
PILE#2					
PILE#3					
PILE#4					
PILE#5					
PILE#6					
PILE#7					
PILE#8					
TOTAL STACKING					
LOCATION	TH-1	TP-6		TP-21	
FIRE HYDRANT PRESSURE					
EQUIPMENTS NOT AVAILABLE					

Appendix – XIII: Format for Operation Desk Log Sheet

STAGE # I / II		CHP : SHIFT LOG				A / B / C				DATE :.....			
Shift Executives:													
Shift Operators:													
Communication register checked		Yes / No				Protection bypass register checked				Yes / No			
01	Running Streams												
02	Bunker Levels	Time	A	B	C	D	E	F	G	H	J	K	
	UNIT- 1 / 3												
	UNIT- 2 / 4												
03	Mills in Service	U # 1/3: A B C D E F G H J K						U # 2/4: A B C D E F G H J K					
04	Clow Flow / Load	U # 1/3						U # 2/4					
05	Belt weigher	BW-1/7		BW-2/8		BW-3/9		BW-4/10		SR-1/3		SR-2/4	
06	Sump Pump Status							SM Status					
07	MD Status							ILMS Status					
08	Equipment Under Permit												
09	Protections Out of Service												
10	Fire zones exceptions												
11	Fire water pressures	TH-1				TP-6				TP-21			
12	TH Position												
13	STACKING	PILE 1 / 5		PILE 2 / 6		PILE 3 / 7		PILE 4 / 8		TOTAL			
	Indian												
	Import												
14	DIRECT BUNKERING	TJH-1		TH-2		WT-1		WT-2		TOTAL			
	Indian Coal												
	Import Coal												
	RECLAIMING	PILE 1 / 5		PILE 2 / 6		PILE 3 / 7		PILE 4 / 8		TOTAL			
	Indian												
	Import												
TOTAL DAY BUNKERING													
15	Blending Status	UNIT # 1		UNIT # 2		UNIT # 3		UNIT # 4		TOTAL			
	Import												
16	Coal Consumption(MT)	UNIT # 1		UNIT # 2		UNIT # 3		UNIT # 4		TOTAL			
	Indian Coal												
	Import Coal												
	TOTAL												
17	Specific Coal consumption												
18	Generation(MU)												
19	Important Communication												

***“HAVING SYSTEMS, PROCESSES AND
PROCEDURES IN PLACE HELPS THE
ORGANISATION TO FUNCTION MORE
SAFELY, PRODUCTIVELY AND
ECONOMICALLY”***